



Human Resources Division
HR Service Center Group/Postdoc Office
Mail Stop P125
Los Alamos, New Mexico 87545
(505) 664-6947, Ext. 05004

October 28, 2010

Dr. Horacio Samaniego
Los Castanos 090
Isla Teja, Valdivia
Region de Los Rios, Chile

Dear Dr. Samaniego,

We are pleased to offer you a **Postdoctoral Research Associate** in the Nuclear & Particle Physics, Astrophysics & Cosmology (T-2) Group. As a Postdoctoral Research Associate, you are provided the opportunity to pursue research as part of ongoing LANL science and engineering programs.

You will be paid for 40 hours per week at an annualized full-time starting salary rate of \$76,300.00, which is based on a standard Laboratory week. This is a two-year, full-time, limited-term appointment. **Your acceptance of this offer is with the understanding that your start date shall be no later than July 2011.** It should be understood that this appointment neither implies nor prohibits subsequent permanent employment with the Laboratory, and does not represent qualifying service under the Laboratory Severance Pay Plan.

In accordance with Laboratory policy all successful applicants must complete a pre-employment drug test. This employment offer is contingent upon meeting requirements set forth by the Laboratory for the pre-employment drug testing. Should you fail to meet these requirements, this offer of employment will be withdrawn. Please refer to the enclosed Pre-Employment Drug Testing Procedures for additional guidance.

This offer is contingent upon the requirements and conditions outlined in this letter. Please do not plan to begin your employment until the contingencies have been met.

Your employment will be effective upon completion of our hiring forms.

Laboratory employees are paid by direct deposit into an account at a financial institution of their choice. You will be required to complete a direct deposit form during your new hire orientation session. You can view this and other new hire forms at <http://www.lanl.gov/orgs/hr/newhires/index.shtml>.

Your employment is contingent upon receipt of the Office of Counterintelligence approval. Your division is in the process of requesting this approval. Please do not report to the Laboratory until you have been officially notified of this approval. Your employment is also contingent upon possession of the appropriate work authorization and/or immigration status. If clarification regarding this issue is necessary, please complete the enclosed VISA Information Form and return it to this office as soon as possible. You may also wish to contact our Immigration Services office at (505) 665-0760.

If you accept this offer, you are required to have a medical evaluation by the Occupational Medicine Group. All information on the employee's health status and history is treated as confidential medical information and is maintained by the Occupational Medicine Group in separate files apart from the employee's personnel records.

You will be eligible for holiday pay, vacation, sick leave, and some insurance benefits if your appointment is 1 year or more. View the Benefits New hire Website at <http://www.lanl.gov/orgs/hr/newhires/index.shtml> for complete information on health and welfare benefits that will be offered to you on your date of hire. You should review the information and print and complete the forms enrolling yourself and eligible family members in the available plans prior to attending orientation. You must provide a social security number for your dependents when you enroll in a medical plan. A Benefits Representative will be on hand to answer specific questions during Orientation.

You will be reimbursed for travel expenses from Chile, by the most direct route to Los Alamos, NM. In addition, you will also be reimbursed for relocation expenses consistent with the Relocation Policy. A copy of the regulations will be sent to you by the Travel Relocation Office. Relocation benefits are extended by the Laboratory with the understanding that should you terminate employment or are terminated prior to 12 months of employment, the Laboratory will require a refund of all or part of these benefits as determined by the hiring organization. Please contact the Laboratory's Relocation Office at (505) 665-4484 ext: 07107 or email relocation@lanl.gov at least two to four weeks prior to your relocation to discuss your relocation plans. You must have met all pre-employment contingencies before the Relocation Office can send any authorizations for household goods shipments.

In addition, should you fail to meet the guidelines set forth by the Laboratory for the pre-employment drug testing you will not be reimbursed for any travel or relocation expenses.

LANL will strive to provide a working environment for you in which the standards for health and safety equal or exceed the highest industrial practices. Our mission includes research and development involving new materials, processes, and technology. All assignments at LANL may be associated directly or indirectly with potential hazards including radioactive and toxic materials. The risks associated with these activities will be maintained at an acceptable level by strict adherence to approved procedures, controls, and technical assistance provided by the Environment, Safety, Health and Quality (ESH&Q) Directorate. It will be your responsibility, with the assistance of your supervisor and ESH&Q Specialist(s), to be aware of risks in your work activities and environment. It is a condition of employment that all safety procedures, rules, and regulations are observed.

As a condition of your employment, you are required to attend and successfully complete one day of General Employee Training (GET). Successful completion of this training may require passing a written exam. Passing the written exam is required for employees whose job assignment requires them to have unescorted access to a radiological controlled area or requires them to work in a nuclear facility. You may also be required to attend and successfully complete additional mandatory training related to the specific duties you will be performing at the Laboratory. During your GET, you will receive a training schedule indicating any additional training that may be required. Failure to successfully complete any required training and/or written exam might adversely affect your Laboratory employment. Please do not report to your work site until the training is successfully completed.

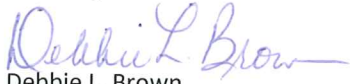
It is a condition of your employment and of your continued employment that you comply with Laboratory policies and procedures.

Pursuant to the Immigration Reform and Control Act (IRCA) of 1986, this offer is also contingent upon your furnishing proof of authorization to work in the United States. On the date you first report to work, you will be required to present either one or a combination of original documents evidencing both your identity and employment eligibility. A list of acceptable documents is enclosed. If you fail to produce these original documents, you will not be hired and you will not be reimbursed for any reimbursable travel or relocation expenses you may have incurred.

Hiring procedures must be completed prior to reporting to your group. New employees are hired on Mondays (Tuesday if Monday is a Laboratory holiday). **Please confirm your date of hire with my office so that we may verify that all necessary pre-employment requirements have been met.** You must report to the Bradbury Science Museum, located on the corner of Central Avenue and 15th Street, on your scheduled arrival date at 8:30 a.m. Should unavoidable circumstances preclude your on-time arrival on Monday, please contact my office. If you do not arrive at the specified time, your hiring date will be delayed or postponed.

Please sign the enclosed copy in the appropriate area, accepting or rejecting this offer and return it to my office. If you accept this offer, please complete the enclosed Supplemental Employment Information and return with your acceptance. If we do not receive a response to this offer by 5 p.m. Mountain Daylight Time, December 28, 2010, this offer becomes null and void. If you have questions regarding this offer, please contact my office. We hope to have you join us in the near future!

Sincerely,



Debbie L. Brown
Human Resource Representative

Cy: T-HR
HR Postdoc file

I accept employment under the conditions and terms as specified herein.

Signature

Date

Full Legal Name (please print)
As it appears on your Passport

Social Security Number

Place of Birth (City and State)

Date of Birth

I plan to begin employment on: _____
Date

I authorize LANL to withhold from my final paycheck any sum that I owe to LANL as reimbursement for monies or benefits provided to me that I have agreed to return to LANL. If the sum of my final paycheck is insufficient to cover the entire amount owed, I agree to pay to LANL, upon demand, the remaining balance.

Signed

Date

I am rejecting this offer.

Signature

Date

Please indicate reason for rejecting offer: _____

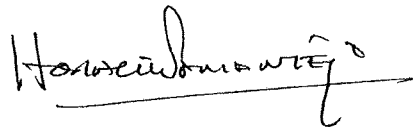
Dynamics of Urban Growth and Energy Systems (Horacio Samaniego)

Recent economic development has significantly increased living standards for the population worldwide. Such prosperity, however, has greatly increased our energy and resource consumption to unprecedented levels exerting important pressures on our natural environment. Most of the growth is in industrial countries where the population has been living in urban areas for some time now. Projections show that roughly 2 billion additional people will migrate to cities in developing nations by about 2030. Life in larger cities offers increased opportunities by fostering the creation of wealth and innovation [1], and at the same time increase energy and resource consumption as well as waste production at a similar rate. Unfortunately, the environmental burden is outpacing much of the environmental and urban planning attempts to manage it.

City growth has recently been characterized in terms of innovation, transport infrastructure, energy and food consumption and distribution [2]. The last three factors, driven by markets that focus on optimization and efficiency, generate economies of scale that will ultimately constrain the size of the city in the long term. Innovation –creation of information, wealth and resources– will foster and accelerate growth to unsustainable levels in the absence of constraints. This is evidenced by the sub- and super-linear scaling relations between the size of cities and their associated infrastructure [3]. These findings show the importance of scaling relations to explain internal dynamics of urban growth. However, they say little about how scaling of urban infrastructures are constrained by the energy and resource throughput of the system. In fact, most predictions have emerged from the analysis of developed countries (USA, Germany, France) and China. It remains to be understood how growth scales in urban systems with different energy generation and supply, economic development, and constraints.

Presuming that the urban infrastructure is a function of the available energy (fuels and electric power), it has important to secure these resources and understand their spatial and temporal flows to growing sustainable and carbon-neutral cities. The scale and complexity of energy systems and their interactions with natural resources makes it very hard to analyze the problem and develop predictions. It is clear from numerous studies, including my work with ecosystems and scaling laws relating road network and commuting times with city size, that detailed spatial and temporal information in models of complex dynamic processes enhances our understanding of the interactions between humans and the environment upon which they depend [4]. My goals during this post-doc are to (i) algorithms for data estimation and validation; (ii) develop integrated models that will help understand the relation between the topological structure of energy systems (networks), food and water availability and the growth of urban infrastructures among countries with similar levels of industrialization; (iii) perform increasingly detailed statistical analysis using these socio-technical networks and relationships and determine factors that constrain or facilitate growth.

Bettencourt at LANL has done pioneering work in applying scaling laws to urban growth and I look forward to collaborating with him. The Global Energy Observatory (GEO) developed by Gupta and Shankar provides a comprehensive framework and database to carry out integrated analysis. GEO will allow me to connect multiple sectors and relate energy production, transmission, conversion and consumption to urban infrastructure growth and surrounding natural resources. The power of the GEO framework lies in its potential to relate energy systems to resources and consumption and how these connections drive and sustain urbanization. The second value of GEO is in understanding how to successfully manage urban environmental impacts with a global perspective. By investigating the distributions of various drivers (and parameters used in models) I plan to extract variables that are common to several regions and those that are region-specific. Based on such an analysis I propose to understand and extract what basic principles underlie these distributions and how those principles relate to specific aspects of urbanization and social life. Finally, I would like to understand the potential environmental impacts of particular configurations of energy systems and develop more robust, efficient and sustainable energy portfolios.



References

- [1] Bretagnolle, A, Pumain, D, & Vacchiani-Marcuzzo, C. (2009) *The Organization of Urban Systems*. pp. 197–220.
- [2] Bettencourt, L. M. A, Lobo, J, Helbing, D, Kuhnert, C, & West, G. B. (2007) *PNAS* **104**, 7301–7306.
- [3] Lammer, S, Gehlsen, B, & Helbing, D. (2006) *Physica A: Statistical Mechanics and its Applications* **363**, 89–95.
- [4] Luck, M, Jenerette, G, Wu, J, & Grimm, N. (2001) *Ecosystems* **4**, 782–796.